Kush Patel

ML Engineer | U.S. Citizen | (916) 291-1781 | patelskush@gmail.com | https://patelskush.com/ | https://github.com/Kush-S-Patel

EDUCATION

University of California, Berkeley Expected May 2027

Master of Information and Data Science

Coursework: DATASCI 201 - Research Design and Applications for Data and Analysis, DATASCI 203 - Statistics for Data Science, DATASCI 205 - Fundamentals of Data Engineering, DATASCI 207 - Applied Machine Learning, DATASCI 221 - Modern Artificial Intelligence (AI) Strategy and Applications, DATASCI 255 - Machine Learning Systems Engineering, DATASCI 266 - Natural Language Processing with Deep Learning (in progress), DATASCI 267 - Fundamentals of Generative Artificial Intelligence (AI) (in progress)

University of California, Berkeley May 2022 - May 2024

Bachelor of Arts in Data Science

Coursework: Data Engineering, Foundations of Data Science, Data Mining & Analytics, Statistical Inference & Decision-Making, Probability Theory & Discrete Mathematics, Intro to Artificial Intelligence (Machine Learning, Neural Networks), Data Structures & Algorithms, Structure and Interpretation of Computer Programs, Advanced iOS Programming (SwiftUI)

EXPERIENCE

Full-Stack Software Engineer — Department of Alcoholic Beverage Control (Information Technology) Jun 2024 - Present

- Design and maintain RESTful APIs in Python (Flask) supporting internal compliance and licensing workflows used by state stakeholders
- Optimize PostgreSQL queries and indexing strategies, improving average endpoint response time by ~20% and reducing redundant database calls
- Build and maintain data ingestion pipelines to process structured and semi-structured data for reporting and anomaly detection
- Containerize services using Docker and implement CI/CD workflows to streamline deployments and reduce manual release overhead

AI Software Engineering Intern — moccet Inc. Apr 2026 - August 2026

- Developed production-ready multi-agent AI workflows using Anthropic-powered prompting systems, improving conversational routing accuracy and reducing manual intervention across AI workflows
- Built and iterated on an internal medical decision-tree (MDT) assistant integrating contextual memory, consult-style reasoning, and structured prompt pipelines across multiple AI agents
- Collaborated directly with the CTO to prototype and deploy AI product features involving prompt engineering, evaluation frameworks, and model orchestration for real-world user interactions
- Designed reusable prompt templates and evaluation logic that improved testing consistency and accelerated AI feature iteration cycles by approximately 30%

Founding Software Engineer Intern — Transnode AI, LLC Apr 2026 - August 2026

- Contributed across the full stack to develop AI-driven platform features using Python, JavaScript, Firebase, and API integrations in a fast-paced startup environment
- Built and shipped internal tooling and automation workflows, including an npm-based developer setup wizard that reduced onboarding and environment setup time by approximately 60%

- Implemented backend optimizations and resolved production bugs, improving application reliability and reducing debugging turnaround time during feature releases
- Worked directly with the founder/CEO on rapid product iteration, technical research, and infrastructure decisions supporting scalable AI product development

Data Analytics Intern — Real Fashion Index Apr 2026 - August 2026

- Monitored website and campaign performance using GA4, Google Search Console, Hotjar, and multi-channel analytics platforms to identify traffic, engagement, and conversion trends
- Built interactive KPI dashboards and executive reporting tools in Looker Studio, improving visibility into SEO, traffic acquisition, and campaign performance metrics
- Analyzed user behavior flows, referral traffic, and high-performing pages to recommend UX and SEO optimizations supporting higher-quality site interactions
- Assisted with frontend and backend analytics implementation tasks, including event tracking and reporting automation through Google Tag Manager workflows

Student Assistant - Software Engineer — Department of Justice (Criminal Justice Information Services) Jul 2020 - Jan 2022

- Developed and maintained full-stack features using React and Python, supporting internal case management and reporting systems
- Built data validation and transformation logic to ensure integrity across multiple government data sources
- Collaborated with cross-functional teams to translate regulatory requirements into scalable back-end workflows
- Implemented authentication and role-based access controls to strengthen application security and compliance

Student Assistant - Data Analytics — Department of Justice (Application Development Bureau) Jun 2023 - Aug 2023

- Analyzed structured datasets using Python (Pandas) to identify patterns and inconsistencies in operational records
- Built regression-based models to explore drivers of anomalous reporting trends, improving internal review prioritization
- Automated recurring Excel-based reporting processes, reducing manual effort and reporting errors
- Presented findings to senior staff, translating statistical insights into clear, actionable recommendations

CERTIFICATIONS, PROJECTS & PRODUCTS

Certifications: CompTIA Security+ (2024)

Shipped Products:
Founder, Educational Nonprofit iOS App — 7000+ downloads, 50000+ impressions
App Store Products — 100000+ purchases, 2000000+ impressions

HSP Lab — Clinical PSG Analysis & ML Decision Support — GitHub 2026

- Built a full-stack clinical decision-support platform for polysomnography (PSG) analysis on the Human Sleep Project dataset (119,000+ recordings from ~90,000 patients across five U.S. clinical centers)
- Developed an interactive study-triage dashboard with AHI, OSA severity, clinical metrics, and review status, integrating automated and human-scored annotations, hypnograms, respiratory event timelines, signal previews, and epoch-level QC tools
- Implemented automated-scoring concordance analysis using Cohen’s kappa, AUROC, and Bland–Altman methods to evaluate model outputs against reference annotations

CareCall — Post-Discharge AI Follow-Up — GitHub 2026

- Built CareCall, a post-discharge follow-up platform that reaches patients by SMS or voice in their preferred language to confirm medication adherence and recovery
- Combined deterministic clinical red-flag tripwires with LLM classification so dangerous symptoms escalate to humans before conversational AI continues
- Designed privacy-preserving architecture with an identity vault and on-device PDF structuring so the LLM never sees raw patient identifiers

EEG Seizure-Marker Proximity Detection — GitHub 2026

- Built a patient-independent seizure detection pipeline on a 10.2 TB, 23,607-recording clinical EEG corpus (4,914 patients), from raw BIDS ingestion through a 244 GB preprocessed window cache with zero patient leakage across train/val/test splits
- Trained an EEG Conformer with focal loss, improving PR-AUC from 0.082 (logistic regression baseline) to 0.46 and ROC-AUC to 0.86 through systematic benchmarking against EEGNet and classical baselines
- Evaluated with clinically relevant metrics (event-level false alarms per 24h, precision at fixed recall) rather than accuracy alone, and ran a documented failure analysis isolating why preictal seizure forecasting stayed near chance

NBA Game Outcome Prediction Model — GitHub 2026 - Present

- Developed an NBA game outcome prediction model achieving 0.75 ROC-AUC by training and benchmarking multiple models (XGBoost, Random Forest, Logistic Regression)
- Engineered advanced team, schedule, and player-level features including custom Elo ratings for teams and players
- Built a full data pipeline by scraping and aggregating historical game data

MyTextJournal — AI-Powered SMS Journaling — App Store 2024 - Present

- Designed, developed, and launched MyTextJournal, an AI-powered journaling platform for SMS/MMS entries with reflective, CBT-informed AI responses (live on the Apple App Store)
- Adapted and fine-tuned a language model for conversational journaling, prioritizing relevant follow-up questions over generic AI advice
- Built SMS/MMS journaling workflows (including photo messages) via messaging infrastructure to lower friction for daily journaling habits

Telco Customer Churn Prediction — GitHub 2025 - Present

- Built and evaluated machine learning models to predict customer churn using structured telecom data
- Achieved strong ROC-AUC performance and improved recall for high-risk customers through threshold tuning
- Interpreted model outputs using SHAP and feature importance analysis to derive actionable business insights

SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS, TypeScript | **Data & ML:** Machine Learning, Large Language Models (LLMs), Prompt Engineering, Multi-Agent Systems, LLM Evaluation, Generative AI, OpenAI API, Anthropic API, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Data Visualization, SEO Analytics | **Databases:** PostgreSQL, Firebase, MySQL, SQLite | **Frameworks:** Flask, SwiftUI, Node.js, React, Express.js, Pandas, Next.js, FastAPI | **Cloud & DevOps:** AWS, GitHub Actions, REST APIs, CI/CD | **Tools:** Google Analytics 4, Looker Studio, Google Tag Manager, Hotjar, Git, GitHub, Semrush, Twilio